
ENTERPRISE TECHNICAL BLUEPRINT

FreightQuote AI

Agentic Maritime Brokerage & Freight Quotation Platform — Development & Implementation Plan

DOCUMENT TYPE	Technical Design & Development Blueprint (implementation-ready)
PROJECT CODE	FQ-AMB-001
DOMAIN	Maritime Logistics / Freight Brokerage / Agentic AI
DURATION	8 Weeks (4 Milestones) + Week 0 Mobilisation
ARCHITECTURE	React SPA • Django REST Core API • FastAPI Agent Service • PostgreSQL + PostGIS + pgvector
AUDIENCE	Engineering Team, Tech Lead, QA, DevOps, Domain SME, Reviewers
STATUS	Baseline v1.0 — approved for build
NOTE	This document contains no source code. It is a specification a team implements from.

Table of Contents

1. Scope, Objectives & Success Criteria

What is being built and how success is measured

2. Solution Architecture Overview

Layered architecture and service decomposition

3. Recommended Technology Stack

Per-layer choices with rationale and alternatives

4. Frontend Blueprint

Folder structure, page inventory, routing, state, conventions

5. Backend Blueprint

Service split, folder structures, API architecture, endpoint catalogue

6. Database Design

Entities, columns, relationships, spatial and vector data

7. Authentication, Authorisation & Roles

Token strategy, RBAC matrix, approval authority

8. AI / ML & Agent Design

Five domain agents, ML models, orchestration, evaluation

9. Third-Party Integrations

Carriers, AIS, weather, customs, notifications

10. End-to-End Project Workflow

Quote lifecycle, state machine, approval rules

11. Development Phases — Step by Step

Week 0 to Week 8, task-level plan with Definition of Done

12. Deployment Architecture

Environments, containers, CI/CD, scaling, observability

13. Non-Functional Requirements, Testing & Security

Performance, QA strategy, compliance

14. Team Structure, Ways of Working & Risk Register

Squads, Git flow, ceremonies, risks

1. Scope, Objectives & Success Criteria

1.1 Problem Restatement

Freight brokerage is a race between accuracy and speed. A customer asks for a price on a container from Nhava Sheva to Rotterdam; the broker opens four spreadsheets, checks two carrier portals, guesses at congestion, half-remembers the customs paperwork for that commodity, applies a margin by instinct and replies six hours later. By then the enquiry may already be quoted elsewhere. The cost of this manual process is threefold: quotes are slow, prices are inconsistent, and compliance exposure is invisible until it becomes a penalty.

FreightQuote AI replaces that process with a network of specialised agents. A shipment request is decomposed across Route, Pricing, Weather, Customs and Margin agents working in parallel; their findings are fused by a Freight Intelligence Engine into a single priced, risk-scored, compliance-checked quotation with an auditable rationale behind every number.

1.2 In Scope

Capability	Description
Shipment request intake	Broker-entered enquiry, customer self-service portal, and REST/EDI ingestion from customer systems.
Route intelligence	Optimal and alternative routings, transit time estimation, carrier comparison, port congestion assessment.
Freight pricing	Cost build-up from base rate plus surcharges, carrier rate comparison, dynamic adjustment by lane and season.
Weather risk	Route-level weather assessment, storm and disruption risk, delay probability, alternative routing advice.
Customs compliance	Import/export rule validation, documentation checklist, HS-code sanity checks, compliance risk flags and readiness report.
Margin optimisation	Cost breakdown, profitability analysis, margin recommendation with floor enforcement, win-probability estimate.
Freight Intelligence Engine	Aggregation of all agent findings, composite shipment risk score, final quotation assembly, agent decision coordination.
Quotation lifecycle	Versioning, approval workflow, customer delivery, validity expiry, acceptance and conversion tracking.
Dashboards	Quotation, Route Intelligence, Customs, Risk and Executive Brokerage dashboards.
Notifications	Quote approval requests, customer notifications, email alerts, Teams/Slack messages, shipment risk alerts.

1.3 Explicitly Out of Scope (Phase 1)

- Direct booking with carriers. The platform produces quotations; booking remains a manual downstream action.
- Automated payment, invoicing or settlement.
- Live vessel tracking as a customer-facing product. AIS data is consumed for congestion and ETA signals only.
- Customs filing or submission to authorities. The system advises on requirements; it does not lodge declarations.
- Air, road and rail multimodal legs beyond simple pre- and on-carriage cost line items.

Domain caveat the team must internalise

Nothing this system outputs is a binding rate, a customs ruling, or legal advice. Every quotation carries a validity window, an assumptions block, and a disclaimer. Every customs output is a *readiness indicator*, reviewed by a human compliance officer before it reaches a customer. Building this expectation into the UI from Milestone 1 is a requirement, not a nicety — a plausible-sounding wrong tariff classification is far more damaging here than a missing feature.

1.4 Success Criteria (contractual, mapped to milestones)

Milestone	Criterion	Measurement Method	Target
M1 / Week 2	Route recommendations generated	200 test lanes across 4 trade regions	≥ 98% return ≥ 2 viable routings
M1 / Week 2	Transit time estimation operational	MAE against historical actuals on a hold-out set	MAE ≤ 2.0 days
M1 / Week 2	Quotation dashboard functional	UAT walkthrough	All list, filter, detail views working
M2 / Week 4	Freight pricing engine operational	Reconciliation against 150 historical quotes	≤ 8% mean absolute deviation on total cost
M2 / Week 4	Margin optimisation functioning	Floor-violation test across all lanes	Zero quotes below configured margin floor
M2 / Week 4	Dynamic quote generation	End-to-end timing test	Quote produced in < 60 s p95
M3 / Week 6	Weather risk analysis operational	Coverage test across all active lanes	100% of routes receive a risk score
M3 / Week 6	Customs validation functioning	50 curated shipment scenarios	≥ 90% documentation checklists correct

M3 / Week 6	Customs validation functioning	30 curated shipment scenarios reviewed by an SME	≥ 90% documentation checklists correct
M3 / Week 6	Shipment risk scoring	Composite score present and explainable	100% of quotes carry a scored, itemised rationale
M4 / Week 8	End-to-end platform deployed	Production release with monitoring	Live, all health checks green
M4 / Week 8	Freight Intelligence Engine operational	Agent orchestration completion rate over 200 runs	≥ 95% complete without error
M4 / Week 8	Executive dashboards available	UAT sign-off by brokerage management	All five dashboards delivered

2. Solution Architecture Overview

2.1 Architectural Style

Three deployable units, same reasoning as any well-scoped agentic platform: keep the transactional system of record separate from the AI compute, and keep long-running work off the request path. Agents here are heavier than in a typical assistant — a single quote may fan out to five external data sources — so the isolation matters more, not less.

Deployable Unit	Technology	Responsibility
Web Client	React + Vite SPA	Broker workbench, customer quote portal, five dashboards, admin console.
Core API	Django + DRF	Identity, RBAC, customers, shipments, quotations, versioning, approval workflow, rate cards, carriers, ports, audit. Owns the system of record and every number that reaches a customer.
Agent Service	FastAPI	Route, Pricing, Weather, Customs and Margin agents plus the Freight Intelligence Engine. Stateless, async, horizontally scalable.
Async Workers	Celery + Redis	Carrier rate refresh, AIS and congestion polling, weather refresh, quote expiry sweeps, notification dispatch, analytics rollups, long-running agent runs.

The single most important architectural rule for this system

The Agent Service *recommends*; the Core API *decides and records*. Margin floors, approval thresholds, rate-card selection and quote validity are business rules and live in Django, enforced deterministically. An agent may suggest a margin of 11%; if the lane floor is 13%, the Core API applies 13% and records that an override was suppressed. Never let a language model be the last thing between a cost calculation and a customer-facing price.

2.2 Layer Model

Layer	Components
Presentation	Broker Workbench • Customer Quote Portal • Route / Customs / Risk / Executive dashboards • Admin console • PDF quotation documents • Email & Teams notifications
API Gateway / Edge	Nginx ingress • TLS termination • rate limiting • request ID injection • WAF
Application (Core)	Auth & RBAC • Shipment & quotation lifecycle • Rate card engine • Margin floor & approval rules • Carrier & port master data • Document generation • Audit
Agent Intelligence	Route Agent • Pricing Agent • Weather Agent • Customs Agent • Margin Agent • Freight Intelligence Engine • Orchestrator • Guardrails
Model & Knowledge	Transit-time model • Congestion forecaster • Rate prediction model • Delay-probability model • Win-probability model • Customs regulation corpus (pgvector)
Integration	Carrier rate APIs • AIS / vessel tracking • Weather providers • Port data • Customs & HS-code references • FX rates • Email / Teams / Slack
Data	PostgreSQL (OLTP) • PostGIS (route geometry, port coordinates) • pgvector (regulation embeddings) • Redis • Object storage (quote PDFs, documents) • Analytics rollups
Observability	Prometheus • Grafana • Loki • Sentry • Langfuse (agent tracing) • structured JSON logs

2.3 Request Flow (happy path)

1. Broker or customer submits a shipment request — origin port, destination port, cargo type, weight/volume, container type, ready date, incoterm. Core API validates against port and commodity master data, persists `shipments`, returns immediately.

2. Core API enqueues `generate_quote`. The requester sees a "quote in progress" state; nothing blocks on agent latency.

3. Agent Service starts a run. **Route Agent** executes first — every downstream agent needs a routing to reason about.

4. Pricing, Weather and **Customs** agents then execute *in parallel* against the candidate routes. This parallelism is the difference between a 20-second quote and a 90-second one.

5. Margin Agent runs last, since it needs the full cost build-up and the risk picture to recommend a margin.

6. Freight Intelligence Engine fuses all findings: selects the recommended routing, assembles the cost breakdown, computes a composite shipment risk score, and writes an itemised rationale.

7. Core API applies deterministic business rules — margin floor, rate-card override, approval threshold — and persists `freight_quotes` version 1 with status `DRAFT` or `PENDING_APPROVAL`.

8. On approval, a PDF quotation is generated, delivered by email and exposed on the customer portal, with a validity expiry timer running. Acceptance, rejection or expiry all feed the win-probability training set.

3. Recommended Technology Stack

3.1 Frontend

Concern	Choice	Rationale / Notes
Framework	React 18 + TypeScript	Strict mode. The domain model (shipment, route, rate, quote version, risk) is complex enough that types are load-bearing, not decoration.
Build tool	Vite 5	Fast HMR, straightforward env handling.
Styling	Tailwind CSS v4 + design tokens	Utility-first with a token layer for the brokerage theme.
Components	shadcn/ui (Radix primitives)	Ownable, accessible, no heavyweight theme lock-in.
Server state	TanStack Query v5	Caching, background refresh, polling for in-progress quote runs.
Client state	Zustand	Session, quote-builder draft state, dashboard filters.
Routing	React Router v6 (data router)	Nested layouts, role guards, separate portal and workbench trees.
Forms	React Hook Form + Zod	The shipment request form has heavy conditional validation (container type by cargo, hazmat fields, incoterm-dependent cost responsibility). Schema-first is the only maintainable approach.
Maps	MapLibre GL (or Mapbox GL) + deck.gl	Great-circle route rendering, port markers, congestion heat layer, vessel positions. MapLibre avoids per-load licensing cost.
Charts	Recharts	Margin distribution, rate trends, on-time performance, revenue forecast.
Tables	TanStack Table	Quote and rate grids need column pinning, grouping and virtualised rows.
HTTP	Axios with interceptors	Token refresh, correlation ID, error normalisation.
Realtime	SSE (primary) / polling fallback	Agent run progress; a quote takes tens of seconds and the user must see which agent is working.
Testing	Vitest + RTL + Playwright	Unit, component, E2E.

3.2 Backend

Concern	Choice	Rationale / Notes
Core framework	Django 5.x + DRF	ORM, migrations, admin, permissions, transactions. Financial calculations need real transactional integrity, not eventual consistency.
Agent framework	FastAPI + Pydantic v2 + Uvicorn	Async matters enormously: four external providers queried concurrently per quote.
Task queue	Celery 5 + Redis	Queues: quotes, market-data, notifications, documents, analytics.
Scheduler	Celery Beat	Carrier rate refresh, AIS/congestion polling, weather refresh, quote expiry sweep, nightly rollups.
Numeric handling	Python Decimal end to end	Never float for money. All monetary columns NUMERIC(14,4); rounding applied only at presentation with an explicit policy.
Currency	Base currency + stored FX rate per quote	The FX rate used is frozen onto the quote. A quote must reproduce identically a month later.
Document generation	WeasyPrint / ReportLab	Branded PDF quotations, customs readiness reports, executive summaries.
API docs	drf-spectacular + FastAPI OpenAPI	Contract published before implementation each milestone.

3.3 Data, AI and Platform

Concern	Choice	Rationale / Notes
Primary database	PostgreSQL 16	Relational integrity for quotes and rates; JSONB for agent payloads.
Spatial	PostGIS extension	Port coordinates, route geometry, great-circle distance, proximity queries, chokepoint intersection for risk.
Vector store	pgvector (HNSW)	Customs regulations, trade documentation guides, commodity restriction notes.
Time series	Partitioned tables (TimescaleDB optional)	Historical freight rates, port waiting hours, weather observations. Monthly partitions are sufficient at Phase 1 volume.
Cache / broker	Redis 7	Celery broker, carrier rate cache (short TTL), weather cache, rate limiting, idempotency keys.
Object storage	S3-compatible	Quote PDFs, customer documents, exported reports. Pre-signed URLs only.
Classical ML	scikit-learn + LightGBM	Transit time, rate prediction, delay probability, win probability. Gradient boosting on tabular logistics data outperforms deep learning here and is far easier to explain to a broker.

Optimisation	NetworkX / OR-Tools	Route graph search over the port-and-service network; constrained margin optimisation.
Time series forecasting	statsmodels / Prophet	Port congestion and seasonal rate trends.
LLM	Provider-abstracted (Claude / GPT-class)	Used for reasoning, summarisation, regulation interpretation and narrative rationale — not for arithmetic.
Agent orchestration	LangGraph (or explicit in-house state machine)	Needs parallel node execution, conditional edges and checkpointing.
Model registry	MLflow (or a versioned S3 artifact store)	Model version recorded on every quote so any historical price can be explained.
LLM observability	Langfuse	Per-agent traces, prompt versions, token cost.
Containers / CI	Docker, Kubernetes, GitHub Actions	Same image promoted across environments.
Monitoring	Prometheus, Grafana, Loki, Sentry	Plus data-freshness dashboards per external provider — stale rate data is the silent failure mode of this system.

4. Frontend Blueprint

4.1 Structural Conventions

Feature-sliced structure. Pages compose; features own logic, data access and domain types. Two distinct application shells exist — the internal **Broker Workbench** and the external **Customer Portal** — and they must not share layout components or navigation config, because the access model differs fundamentally and accidental leakage of internal cost data into the portal is the failure nobody recovers from.

- Any component displaying cost, margin or carrier buy-rate lives under `features/pricing/internal/` and is never imported by a portal page. This is enforced with an ESLint import-boundary rule, not by convention alone.
- Money is rendered exclusively through a shared `<Money />` component that takes a currency and applies the project rounding policy. No ad-hoc `toFixed(2)` anywhere.
- `components/ui/` is domain-free. If it mentions "quote" or "shipment", it belongs in a feature.

4.2 Folder Structure

```
freightquote-web/
├── public/
│   ├── favicon.svg
│   └── map-styles/                # MapLibre style JSON
├── src/
│   ├── app/
│   │   ├── App.tsx
│   │   ├── router.tsx            # two route trees: /portal and /app
│   │   └── providers/
│   │       ├── QueryProvider.tsx
│   │       ├── AuthProvider.tsx
│   │       ├── CurrencyProvider.tsx  # base currency + FX display context
│   │       ├── ThemeProvider.tsx
│   │       └── ToastProvider.tsx
│   │   └── layouts/
│   │       ├── AuthLayout.tsx
│   │       ├── WorkbenchLayout.tsx  # internal: sidebar + topbar + agent status rail
│   │       ├── PortalLayout.tsx     # external customer shell
│   │       └── AdminLayout.tsx
│   ├── pages/
│   │   ├── auth/
│   │   │   ├── LoginPage.tsx
│   │   │   └── ResetPasswordPage.tsx
│   │   ├── portal/                # CUSTOMER-FACING ONLY - no cost data
│   │   │   ├── RequestQuotePage.tsx
│   │   │   ├── MyQuotesPage.tsx
│   │   │   ├── QuoteViewPage.tsx
│   │   │   └── DocumentsPage.tsx
│   │   ├── dashboard/
│   │   │   └── BrokerHomePage.tsx
│   │   ├── shipments/
│   │   │   ├── ShipmentListPage.tsx
│   │   │   ├── ShipmentDetailPage.tsx
│   │   │   └── NewShipmentPage.tsx
│   │   ├── quotes/
│   │   │   ├── QuoteListPage.tsx
│   │   │   ├── QuoteBuilderPage.tsx
│   │   │   ├── QuoteDetailPage.tsx
│   │   │   ├── QuoteComparePage.tsx
│   │   │   └── ApprovalQueuePage.tsx
│   │   ├── routes/
│   │   │   ├── RouteIntelligencePage.tsx
│   │   │   ├── RouteDetailPage.tsx
│   │   │   ├── PortCongestionPage.tsx
│   │   │   └── CarrierComparisonPage.tsx
│   │   ├── pricing/
│   │   │   ├── PricingWorkbenchPage.tsx
│   │   │   ├── RateCardPage.tsx
│   │   │   ├── CostBreakdownPage.tsx
│   │   │   └── MarginAnalysisPage.tsx
│   │   ├── risk/
│   │   │   ├── RiskDashboardPage.tsx
│   │   │   ├── WeatherRiskPage.tsx
│   │   │   └── AlertsPage.tsx
│   │   ├── customs/
│   │   │   ├── CustomsDashboardPage.tsx
│   │   │   ├── ComplianceCheckPage.tsx
│   │   │   ├── DocumentChecklistPage.tsx
│   │   │   └── RegulationLibraryPage.tsx
│   │   ├── agents/
│   │   │   ├── AgentMonitorPage.tsx
│   │   │   └── AgentRunDetailPage.tsx
│   │   ├── executive/
│   │   │   ├── ExecutiveDashboardPage.tsx
│   │   │   ├── RevenueForecastPage.tsx
│   │   │   └── LanePerformancePage.tsx
│   │   ├── masterdata/
│   │   │   ├── CustomersPage.tsx
│   │   │   ├── CarriersPage.tsx
│   │   │   ├── PortsPage.tsx
│   │   │   └── CommoditiesPage.tsx
│   └── admin/
```

```

├── UsersPage.tsx
├── RolesPage.tsx
├── MarginPolicyPage.tsx
├── ApprovalRulesPage.tsx
├── IntegrationsPage.tsx
├── AgentSettingsPage.tsx
├── AuditLogPage.tsx
├── errors/
├── features/
├── auth/
├── shipments/
├──   api/{shipmentApi,shipmentKeys}.ts
├──   hooks/{useShipments,useCreateShipment}.ts
├──   components/{ShipmentForm,CargoDetailsFields,
├──               IncotermSelector,HazmatPanel,ShipmentTimeline}.tsx
├──   schemas/shipmentSchema.ts
├──   types.ts
├── quotes/
├──   api/quoteApi.ts
├──   hooks/{useQuote,useQuoteRun,useApproveQuote}.ts
├──   components/
├──   │   QuoteSummaryCard.tsx
├──   │   QuoteVersionTimeline.tsx
├──   │   QuoteStatusBadge.tsx
├──   │   ValidityCountdown.tsx
├──   │   AssumptionsPanel.tsx
├──   │   QuotePdfPreview.tsx
├──   types.ts
├── routing/
├──   api/routeApi.ts
├──   components/{RouteMap,RouteOptionCard,TransitTimeBar,
├──               PortCongestionGauge,CarrierCompareTable,
├──               ChokepointBadge}.tsx
├──   types.ts
├── pricing/
├──   internal/ # cost + margin - NEVER imported by portal
├──   │   CostBreakdownTable.tsx
├──   │   SurchargeEditor.tsx
├──   │   MarginSlider.tsx
├──   │   MarginFloorWarning.tsx
├──   │   WinProbabilityMeter.tsx
├──   api/pricingApi.ts
├──   components/{SellRateSummary,CurrencyToggle}.tsx
├── weather/
├──   components/{WeatherRiskMap,StormAlertCard,
├──               DelayProbabilityChart,RouteWeatherTimeline}.tsx
├── customs/
├──   components/{ComplianceChecklist,DocumentRequirementCard,
├──               HsCodeLookup,RegulationCitation,
├──               ReadinessScore}.tsx
├── agents/
├──   hooks/useAgentRunStream.ts
├──   components/{AgentPipelineRail,AgentStatusChip,
├──               AgentStepCard,RationaleTrace}.tsx
├── analytics/
├──   components/{KpiTile,MarginByLaneChart,RevenueForecastChart,
├──               ConversionFunnel,AgentPerformancePanel}.tsx
├── admin/
├── components/
├──   ui/ # Button, Input, Select, Dialog, Tabs, Toast...
├──   common/ # PageHeader, EmptyState, ErrorBoundary,
├──           # ConfirmDialog, Money, DateRange
├──   data/ # DataTable, Pagination, ExportButton
├──   maps/ # MapContainer, PortMarker, RouteLine, HeatLayer
├──   charts/
├── hooks/ # useDebounce, usePagination, useCountdown
├── lib/
├──   axios.ts
├──   queryClient.ts
├──   rbac.ts
├──   sse.ts
├──   money.ts # formatting + rounding policy (single source)
├──   units.ts # TEU, CBM, kg/lb, nautical miles
├──   geo.ts # great-circle helpers for map rendering
├──   constants.ts # enums mirrored from backend
├── store/{authStore,quoteBuilderStore,filterStore,uiStore}.ts
├── types/ # api.d.ts (generated from OpenAPI), domain.ts
├── config/ # env.ts, routes.ts, navigation.ts, permissions.ts
├── styles/
├── main.tsx
├── tests/{unit,component,e2e}/
├── .env.example
├── vite.config.ts
├── tailwind.config.ts
├── eslint.config.js # includes import-boundary rule for pricing/internal

```

4.3 Page & Route Inventory

Route	Page	Allowed Roles	Purpose & Key Elements
/login	LoginPage	Public	Credential or SSO sign-in; customers and staff use the same endpoint, different landing routes.

/portal/request	RequestQuotePage	Customer	Self-service enquiry: ports, cargo, containers, ready date, incoterm. Shows indicative transit time as the form is completed.
/portal/quotes	MyQuotesPage	Customer	Quote list with status and validity countdown.
/portal/quotes/:id	QuoteViewPage	Customer	Sell rate, routing, transit time, inclusions/exclusions, assumptions, accept/reject actions. No cost or margin data.
/app	BrokerHomePage	Broker+	My open enquiries, quotes awaiting approval, expiring today, high-risk shipments, conversion rate.
/app/shipments/new	NewShipmentPage	Broker+	Full enquiry form with conditional cargo and hazmat sections.
/app/shipments	ShipmentListPage	Broker+	Filterable table by lane, customer, status, date.
/app/shipments/:id	ShipmentDetailPage	Broker+	Shipment facts, linked quotes, agent run history, alerts.
/app/quotes	QuoteListPage	Broker+	All quotes with margin, status, validity, win probability columns.
/app/quotes/builder/:shipmentId	QuoteBuilderPage	Broker+	The core working screen: route options, cost build-up, surcharge editor, margin slider with floor warning, live sell rate, agent rationale rail, generate/save/submit actions.
/app/quotes/:id	QuoteDetailPage	Broker+	Version timeline, full breakdown, risk score with itemised rationale, customs readiness, PDF preview, send to customer.
/app/quotes/compare	QuoteComparePage	Broker+	Side-by-side comparison of route/carrier options for the same shipment.
/app/quotes/approvals	ApprovalQueuePage	Senior Broker, Pricing Mgr	Quotes breaching margin floor or discount authority; approve, reject with reason, request revision.
/app/routes	RouteIntelligencePage	Broker+, Ops	Global route map, lane on-time performance, transit analytics, route efficiency score.
/app/routes/:id	RouteDetailPage	Broker+, Ops	Leg-by-leg detail, transshipment ports, chokepoints, historical variance.
/app/routes/congestion	PortCongestionPage	Broker+, Ops	Waiting hours by port, berthing status counts, congestion trend and forecast.
/app/routes/carriers	CarrierComparisonPage	Broker+	Carrier reliability, transit, service frequency, rate positioning.
/app/pricing	PricingWorkbenchPage	Broker+, Pricing Mgr	Lane rate explorer, historical rate trend, seasonality view.
/app/pricing/rate-cards	RateCardPage	Pricing Mgr, Admin	Contract and spot rate cards, validity periods, surcharge definitions, upload/import.
/app/pricing/margins	MarginAnalysisPage	Pricing Mgr, Exec	Margin distribution by trade lane, customer and carrier; outlier detection.
/app/risk	RiskDashboardPage	Broker+, Ops, Exec	Risk distribution by region, high-risk shipment list, composite score drivers.
/app/risk/weather	WeatherRiskPage	Broker+, Ops	Weather overlay map, active storm alerts, delay probability by lane.
/app/risk/alerts	AlertsPage	Broker+, Ops	Alert inbox: severity, affected shipments, acknowledge and resolve.
/app/customs	CustomsDashboardPage	Compliance, Broker+	Compliance rate, pending documentation, flagged shipments by country.
/app/customs/check/:shipmentId	ComplianceCheckPage	Compliance, Broker+	Requirement checklist with regulation citations, readiness score, officer sign-off action.
/app/customs/regulations	RegulationLibraryPage	Compliance, Admin	Regulation corpus, semantic search, ingestion status, source freshness.
/app/agents	AgentMonitorPage	Ops, Admin	Live agent runs, per-agent success rate, latency, failures.
/app/agents/runs/:id	AgentRunDetailPage	Ops, Admin	Full trace: each agent's inputs, outputs, data sources used, model versions, tokens, latency.
/app/executive	ExecutiveDashboardPage	Executive, Admin	Brokerage revenue, quote conversion rate, margin improvement, shipment success rate, agent performance overview.
/app/executive/forecast	RevenueForecastPage	Executive, Admin	Pipeline-weighted revenue forecast by lane and period.
/app/executive/lanes	LanePerformancePage	Executive, Pricing Mgr	Volume, margin and win rate per trade lane; underperforming lane flags.
/app/masterdata/*	Customers / Carriers / Ports /	Ops, Admin	Master data CRUD with import; ports carry coordinates,

	Commodities		UN/LOCODE and country.
/app/admin/margin-policy	MarginPolicyPage	Admin, Pricing Mgr	Margin floors and targets per lane, customer tier and cargo type.
/app/admin/approval-rules	ApprovalRulesPage	Admin	Thresholds that trigger approval, approver role per threshold band.
/app/admin/agent-settings	AgentSettingsPage	Admin	Enable/disable agents, timeouts, data-source priority, confidence thresholds, model selection.
/app/admin/integrations	IntegrationsPage	Admin	Provider health, last successful sync, data-freshness indicators, credentials test.
/app/admin/audit	AuditLogPage	Admin	Append-only log; every quote price change and approval is here.

5. Backend Blueprint

5.1 Core API — Folder Structure (Django + DRF)

Same internal contract per app: **models** persist, **selectors** read, **services** write and enforce business rules, **views** stay thin. In this system the services layer carries unusual weight — margin floors, approval routing and quote versioning all live there, and they are the rules an auditor will ask about.

```
freightquote-core/
├── config/
│   ├── settings/{base,local,staging,production}.py
│   ├── urls.py
│   ├── celery.py
│   ├── asgi.py
│   └── wsgi.py
├── apps/
│   ├── accounts/
│   │   ├── models.py                # User, Role, Organization, CustomerContact
│   │   ├── serializers.py
│   │   ├── views.py                # login, refresh, me, users, roles
│   │   ├── services.py
│   │   ├── permissions.py          # IsBroker, IsPricingManager, IsCustomerOfOrg
│   │   └── tests/
│   ├── masterdata/
│   │   ├── models.py                # Port (PointField), Carrier, Service,
│   │   │                           # Commodity, ContainerType, Currency, FxRate
│   │   ├── services.py              # import_ports, refresh_fx
│   │   ├── views.py
│   │   └── tests/
│   ├── customers/
│   │   ├── models.py                # Customer, CustomerTier, CreditProfile,
│   │   │                           # CustomerContact
│   │   ├── services.py
│   │   ├── views.py
│   ├── shipments/
│   │   ├── models.py                # Shipment, CargoDetail, ContainerRequirement,
│   │   │                           # ShipmentStatusHistory
│   │   ├── serializers.py
│   │   ├── services.py              # create_shipment, validate_lane, request_quote
│   │   ├── selectors.py
│   │   ├── state_machine.py
│   │   ├── filters.py
│   │   ├── tasks.py                # generate_quote
│   │   └── tests/
│   ├── routing/
│   │   ├── models.py                # Route, RouteLeg, TransshipmentPoint,
│   │   │                           # PortCongestionSnapshot, RoutePerformance
│   │   ├── services.py              # persist_route_options, select_recommended
│   │   ├── selectors.py              # lane_performance, congestion_trend
│   │   └── views.py
│   ├── pricing/
│   │   ├── models.py                # RateCard, RateCardLine, Surcharge,
│   │   │                           # CarrierRate, CostComponent, RateHistory
│   │   ├── services.py              # build_cost_breakdown, apply_surcharges,
│   │   │                           # resolve_applicable_rate_card
│   │   ├── selectors.py
│   │   ├── importers.py              # rate card CSV/XLSX ingestion + validation
│   │   └── views.py
│   ├── quotes/
│   │   ├── models.py                # FreightQuote, QuoteVersion, QuoteLineItem,
│   │   │                           # QuoteApproval, QuoteDocument, QuoteEvent
│   │   ├── serializers.py
│   │   ├── services.py              # assemble_quote, apply_margin_policy,
│   │   │                           # route for approval, issue, accept, expire
│   │   ├── margin_policy.py          # DETERMINISTIC floor + target resolution
│   │   ├── approval_rules.py        # threshold -> approver role resolution
│   │   ├── state_machine.py
│   │   ├── documents.py              # PDF generation
│   │   ├── tasks.py                  # expiry_sweep, send_quote
│   │   └── tests/
│   ├── risk/
│   │   ├── models.py                # WeatherRisk, ShipmentRiskScore, RiskFactor,
│   │   │                           # Alert, AlertSubscription
│   │   ├── services.py              # compute_composite_risk, raise_alert
│   │   └── views.py
│   ├── customs/
│   │   ├── models.py                # CustomsRequirement, DocumentRequirement,
│   │   │                           # ComplianceCheck, Regulation, RegulationChunk,
│   │   │                           # HsCode, ComplianceSignoff
│   │   ├── services.py              # run_compliance_check, record_signoff
│   │   ├── tasks.py                  # ingest_regulations, reembed
│   │   └── views.py
│   ├── agents/
│   │   ├── models.py                # AgentRun, AgentStep, AgentConfig,
│   │   │                           # DataSourceSnapshot
│   │   ├── services.py              # start_run, record_step, finalize_run
│   │   ├── views.py                  # list, detail, SSE stream, retry
│   │   ├── tasks.py
│   └── integrations/
│       ├── carriers/{client,mapper,tasks,models}.py
│       ├── ais/{client,tasks}.py     # vessel positions, port calls
│       ├── weather/{client,tasks}.py
│       └── portdata/{client,tasks}.py # congestion, waiting hours
```

```

├── customs_ref/{client,tasks}.py
├── fx/{client,tasks}.py
├── messaging/{email,teams,slack}.py
├── models.py                # IntegrationConfig, SyncLog, DataFreshness
├── notifications/
│   ├── models.py           # Notification, NotificationPreference,
│   │                       # NotificationTemplate
│   ├── dispatcher.py
│   └── tasks.py
├── analytics/
│   ├── models.py           # DailyQuoteRollup, LanePerformanceRollup,
│   │                       # AgentPerformanceRollup, RevenueForecast
│   ├── aggregators.py
│   ├── exporters.py
│   ├── views.py
│   └── tasks.py
├── audit/
│   ├── models.py           # ActivityLog (append-only)
│   ├── middleware.py
│   └── views.py
├── core/
│   ├── permissions.py
│   ├── pagination.py
│   ├── exceptions.py
│   ├── middleware.py
│   ├── money.py            # Decimal helpers, rounding policy, FX conversion
│   ├── units.py            # TEU/CBM/weight conversions
│   ├── enums.py
│   └── validators.py
├── services/
│   ├── agent_client.py     # typed HTTP client for the FastAPI service
│   ├── storage.py
│   └── cache.py
├── requirements/{base,local,production}.txt
├── docker/
├── manage.py
└── pytest.ini

```

5.2 Agent Service — Folder Structure (FastAPI)

```

freightquote-agents/
├── app/
│   ├── main.py
│   ├── api/
│   │   └── v1/
│   │       ├── router.py
│   │       └── routes/
│   │           ├── health.py
│   │           ├── quote.py           # full orchestrated run
│   │           ├── route.py          # Route Agent standalone
│   │           ├── pricing.py
│   │           ├── weather.py
│   │           ├── customs.py
│   │           ├── margin.py
│   │           └── intelligence.py    # fusion / risk scoring
│   ├── core/
│   │   ├── config.py
│   │   ├── security.py              # service-to-service auth
│   │   ├── logging.py
│   │   └── errors.py
│   ├── agents/
│   │   ├── base.py                 # Agent protocol: run(state) -> state
│   │   ├── state.py                # typed shared QuoteState
│   │   ├── orchestrator.py         # graph: route -> [pricing|weather|customs]
│   │   │   └── -> margin -> intelligence
│   │   ├── route_agent.py
│   │   ├── pricing_agent.py
│   │   ├── weather_agent.py
│   │   ├── customs_agent.py
│   │   ├── margin_agent.py
│   │   ├── intelligence_engine.py  # fusion, composite risk, rationale writer
│   │   └── tools/
│   │       ├── port_graph.py       # port + service network queries
│   │       ├── carrier_rates.py
│   │       ├── weather_lookup.py
│   │       ├── congestion_lookup.py
│   │       ├── regulation_search.py # RAG over customs corpus
│   │       └── historical_quotes.py
│   ├── ml/
│   │   ├── transit_time/{features,train,predict}.py
│   │   ├── congestion_forecast/{train,predict}.py
│   │   ├── rate_prediction/{features,train,predict}.py
│   │   ├── delay_probability/{train,predict}.py
│   │   ├── win_probability/{features,train,predict}.py
│   │   ├── registry.py             # version tags, load by tag
│   │   └── artifacts/              # gitignored, pulled from model store
│   ├── optimization/
│   │   ├── route_search.py         # graph search over the service network
│   │   ├── cost_model.py           # deterministic cost assembly
│   │   └── margin_optimizer.py     # constrained optimisation
│   ├── rag/
│   │   ├── ingestion.py            # regulation and trade-doc loaders
│   │   ├── chunking.py
│   │   ├── embedding.py
│   │   ├── vectorstore.py          # pgvector adapter
│   │   ├── retriever.py            # hybrid dense + lexical
│   │   └── reranker.py
│   ├── llm/
│   │   ├── base.py                 # LLMProvider protocol
│   │   ├── providers/
│   │   ├── router.py
│   │   └── token_guard.py
│   ├── prompts/
│   │   ├── registry.py
│   │   ├── route_reasoning.v2.yaml
│   │   ├── customs_interpretation.v3.yaml
│   │   ├── risk_rationale.v2.yaml
│   │   └── quote_narrative.v1.yaml
│   ├── guardrails/
│   │   ├── numeric_guard.py        # LLM output NEVER used for arithmetic
│   │   ├── citation_enforcer.py    # customs claims must cite a regulation
│   │   ├── staleness_guard.py       # reject decisions on stale market data
│   │   └── output_validator.py
│   ├── schemas/
│   ├── evaluation/
│   │   ├── datasets/
│   │   ├── metrics.py
│   │   ├── run_eval.py
│   │   └── reports/
│   └── observability/{tracing,metrics}.py
├── tests/{unit,integration,eval}/
├── notebooks/
├── requirements.txt
└── Dockerfile

```

5.3 API Architecture & Conventions

Concern

Standard

Base path	/api/v1/ on Core; /v1/ on the Agent Service (internal network only).
Envelopes	Success { success, data, meta }; error { success:false, error:{ code, message, details[], request_id } } with stable codes such as QUOTE_BELOW_MARGIN_FLOOR, RATE_CARD_EXPIRED, PORT_PAIR_UNSERVICED.
Monetary fields	Always transmitted as a string decimal plus an ISO currency code, never a float. Every priced response includes the fx_rate and fx_rate_date used.
Pagination	Cursor for quotes, shipments and audit; page-number for master data. Default 25, max 100.
Idempotency	Idempotency-Key mandatory on shipment creation, quote issuance and all outbound carrier calls.
Long-running work	Quote generation returns 202 with a run_id; the client polls or subscribes to SSE. Typical run 20-60 s.
Data freshness	Every response sourced from external market data carries data_as_of timestamps per source. The UI must display these — a broker quoting from three-day-old rates needs to know.
Correlation	X-Request-ID propagated edge → Core → Celery → Agent Service → provider calls → LLM traces.
Rate limits	Customer portal 60/min; internal 200/min; quote generation 20/hour/user to bound external API cost.

5.3.1 Core API Endpoint Catalogue

Method	Endpoint	Roles	Description
POST	/auth/login, /auth/refresh, /auth/logout	Public / Any	Token issue, rotation, revocation.
GET	/auth/me	Any	Profile, roles, organisation, effective permissions.
POST	/shipments	Broker+, Customer	Create enquiry; validates port pair, commodity, container type.
GET	/shipments	Scoped	List; customers see only their organisation's.
GET	/shipments/{id}	Scoped	Detail with linked quotes and runs.
POST	/shipments/{id}/generate-quote	Broker+	202; starts the orchestrated agent run.
POST	/shipments/ingest	Service token	Customer-system / EDI ingestion.
GET	/routes/options?shipment_id=	Broker+	Candidate routings with transit, carrier, congestion and risk.
GET	/routes/{id}	Broker+	Leg detail, transshipments, geometry for the map.
GET	/routes/performance	Broker+, Exec	On-time performance by lane and carrier.
GET	/ports/congestion	Broker+, Ops	Waiting hours, berthing counts, trend and forecast.
GET	/carriers/compare?lane=	Broker+	Reliability, frequency, transit, rate positioning.
GET	/pricing/rate-cards	Pricing Mgr, Admin	List with validity windows.
POST	/pricing/rate-cards/import	Pricing Mgr, Admin	CSV/XLSX import with validation report before commit.
GET	/pricing/cost-breakdown?shipment_id=	Broker+	Itemised buy-side cost build-up.
GET	/pricing/rate-history?lane=	Broker+, Pricing Mgr	Historical rate series for the lane.
POST	/quotes	Broker+	Create quote version from a shipment and selected route.
GET	/quotes	Scoped	List with margin, status, validity, win probability.
GET	/quotes/{id}	Scoped	Full detail; cost and margin fields stripped for Customer role at the serializer level.
POST	/quotes/{id}/versions	Broker+	New version; previous version becomes immutable.
PATCH	/quotes/{id}/margin	Broker+	Adjust margin; 409 QUOTE_BELOW_MARGIN_FLOOR if below floor without authority.
POST	/quotes/{id}/submit-approval	Broker+	Route to the approver resolved by threshold rules.
POST	/quotes/{id}/approve	Senior Broker, Pricing Mgr	Approve with optional comment.
POST	/quotes/{id}/reject	Senior Broker, Pricing Mgr	Reject with mandatory reason.
POST	/quotes/{id}/issue	Broker+	Generate PDF, email customer, start validity timer.
POST	/quotes/{id}/accept	Customer	Customer acceptance; records conversion.
POST	/quotes/{id}/decline	Customer	Decline with optional reason — a valuable win/loss signal.
GET	/quotes/{id}/document	Scoped	Pre-signed PDF URL.
GET	/quotes/approvals/queue	Approvers	Pending approvals with breach reason.
GET	/risk/shipments/{id}	Broker+, Ops	Composite risk score with itemised factor contributions.
GET	/risk/weather?route_id=	Broker+, Ops	Route weather assessment and delay probability.
GET/POST	/risk/alerts	Broker+, Ops	Alert list; acknowledge and resolve.

POST	/customs/check	Compliance, Broker+	Run compliance validation for a shipment.
GET	/customs/requirements?shipment_id=	Compliance, Broker+	Documentation checklist with regulation citations.
POST	/customs/signoff/{check_id}	Compliance	Human sign-off before customer delivery.
POST	/customs/regulations/ingest	Compliance, Admin	Ingest regulation documents into the corpus.
GET	/agents/runs, /agents/runs/{id}	Ops, Admin	Run list and full trace.
GET	/agents/runs/{id}/stream	Broker+, Ops	SSE progress for the quote builder.
POST	/agents/runs/{id}/retry	Ops, Admin	Resume from last checkpoint.
GET	/analytics/kpi, /analytics/lanes, /analytics/forecast, /analytics/agents	Exec, Pricing Mgr, Admin	Dashboard data sources.
POST	/analytics/reports/export	Exec, Admin	Queue CSV/PDF export.
GET/PUT	/admin/margin-policies, /admin/approval-rules, /admin/agent-settings	Admin	Business rule configuration.
GET	/integrations/health	Admin, Ops	Per-provider status and data freshness.
GET	/audit/logs	Admin	Append-only activity log.

5.3.2 Agent Service Endpoint Catalogue (internal)

Method	Endpoint	Input	Output
POST	/v1/quote/run	shipment payload, customer tier, options	run_id (202)
GET	/v1/quote/run/{id}	—	status, per-agent results, fused quote proposal, risk score, rationale
POST	/v1/route/options	origin, destination, ready date, container type	ranked routings: legs, carrier, transit days, congestion, confidence
POST	/v1/pricing/estimate	route, cargo, container count, date	cost components, base rate, surcharges, predicted market rate, confidence
POST	/v1/weather/assess	route geometry, departure window	risk score, storm alerts, delay probability, alternative advice
POST	/v1/customs/validate	origin/destination country, HS code, commodity, incoterm	requirements[], documents[], risk flags[], citations[], readiness score
POST	/v1/margin/optimize	cost total, lane, customer tier, risk score, competitive context	recommended margin, floor check, win probability, sensitivity curve
POST	/v1/intelligence/fuse	all agent outputs	recommended routing, composite risk, itemised rationale, confidence
POST	/v1/regulations/search	query, country filters, top_k	ranked regulation chunks with citations
GET	/v1/models	—	active model and prompt versions per agent

6. Database Design

6.1 Design Principles

- UUID primary keys on all business tables.
- All monetary values `NUMERIC(14,4)` with an explicit currency column. Never float, never a single implicit currency.
- **Quote versions are immutable.** A price change creates a new version; the previous row is never updated. This is the audit backbone of the whole system.
- Every quote freezes the inputs that produced it: rate card ID, FX rate, model versions, data-source timestamps. A quote must be fully reproducible months later without depending on current market data.
- Spatial data in PostGIS: ports as `POINT(4326)`, route geometry as `LINESTRING`, enabling distance, proximity and chokepoint-intersection queries.
- Agent outputs stored as `JSONB` alongside the structured columns the business logic actually reads.
- Time-series tables (rate history, congestion snapshots, weather observations) partitioned monthly with a retention policy.

6.2 Core Entities

shipments

Column	Type	Notes
id	UUID PK	
reference	VARCHAR(20) UNIQUE	Human-facing, e.g. SHP-2026-01842.
customer_id	UUID FK → customers	Indexed.
created_by_id	UUID FK → users	Broker or customer contact.
origin_port_id	UUID FK → ports	Indexed with destination as a composite lane index.
destination_port_id	UUID FK → ports	
commodity_id	UUID FK → commodities	
hs_code	VARCHAR(12)	Nullable; drives customs validation.
cargo_type	VARCHAR(25)	FCL / LCL / BREAKBULK / REEFER / HAZMAT
container_type	VARCHAR(15)	20GP / 40GP / 40HC / 40RF / etc.
container_count	SMALLINT	
gross_weight_kg	NUMERIC(12,3)	
volume_cbm	NUMERIC(12,3)	
is_hazmat	BOOLEAN	Triggers additional customs and carrier constraints.
imo_class / un_number	VARCHAR(10)	Required when hazmat.
incoterm	VARCHAR(3)	EXW/FOB/CIF/DAP/DDP — determines which cost legs are in scope.
ready_date	DATE	
required_delivery_date	DATE NULL	Drives urgency and route selection weighting.
status	VARCHAR(25)	See state machine, section 10.2.
special_requirements	TEXT	Free text passed to the agents as context.

freight_quotes & quote_versions

Column	Type	Notes
id	UUID PK	On freight_quotes; the logical quote.
quote_number	VARCHAR(20) UNIQUE	e.g. QT-2026-00934.
shipment_id	UUID FK → shipments	
current_version	SMALLINT	Pointer to the live version.
status	VARCHAR(25)	DRAFT / PENDING_APPROVAL / APPROVED / ISSUED / ACCEPTED / DECLINED / EXPIRED / SUPERSEDED
quote_versions — one immutable row per version		
version	SMALLINT	Unique with quote_id.
route_id	UUID FK → routes	Selected routing.
carrier_id	UUID FK → carriers	

currency	CHAR(3)	Quote currency.
fx_rate / fx_rate_date	NUMERIC(12,6) / DATE	Frozen. Reproducibility depends on this.
total_cost	NUMERIC(14,4)	Buy side. Never exposed to Customer role.
margin_amount / margin_pct	NUMERIC(14,4) / NUMERIC(6,3)	Internal only.
final_quote	NUMERIC(14,4)	Sell rate shown to the customer.
transit_days	SMALLINT	
risk_score	NUMERIC(5,4)	Composite from the Intelligence Engine.
win_probability	NUMERIC(5,4)	Model estimate at issuance.
valid_until	TIMESTAMP TZ	Expiry sweep acts on this.
rate_card_id	UUID FK NULL	Frozen input.
agent_run_id	UUID FK → agent_runs	Traceability from price to reasoning.
model_versions	JSONB	Which model produced each estimate.
data_as_of	JSONB	Per-source freshness timestamps.
assumptions	JSONB	Rendered into the PDF; the legal shield.
rationale	JSONB	Itemised explanation per component.
created_by_id / created_at	UUID / TIMESTAMP TZ	Immutable after insert.

Indexes: (shipment_id, version DESC); (status, valid_until) for the expiry sweep; (origin_port_id, destination_port_id, created_at DESC) on the joined lane view for rate history; partial index on status='PENDING_APPROVAL'.

Remaining tables

Table	Key columns
users	id, email, full_name, password_hash, role, organization_id, customer_id (nullable, for portal users), is_active, last_login_at
customers	id, company_name, country, tier (STRATEGIC/KEY/STANDARD/NEW), credit_status, default_currency, payment_terms, account_manager_id
customer_contacts	id, customer_id, name, email, phone, is_primary
ports	id, un_locode UNIQUE, name, country, region, coordinates GEOGRAPHY(POINT,4326), timezone, is_transshipment_hub
carriers	id, name, scac_code, service_type, reliability_score, on_time_pct, is_active
carrier_services	id, carrier_id, service_name, port_rotation JSONB, frequency_days, is_active
routes	id, shipment_id, route_name, origin_port_id, destination_port_id, carrier_id, transit_days, distance_nm, geometry GEOGRAPHY(LINESTRING,4326), route_risk_score, congestion_score, is_recommended, rank, agent_confidence
route_legs	id, route_id, leg_index, from_port_id, to_port_id, carrier_id, vessel_name, transit_days, is_transshipment
port_congestion_snapshots	id, port_id, snapshot_at, avg_waiting_hours, vessels_waiting, vessels_berthing, congestion_index, source (partitioned monthly)
rate_cards	id, name, carrier_id, type (CONTRACT/SPOT/TARIFF), currency, valid_from, valid_to, uploaded_by, is_active
rate_card_lines	id, rate_card_id, origin_port_id, destination_port_id, container_type, base_rate, currency, min_charge
surcharges	id, code (BAF/CAF/THC/ISPS/PSS/LSS), name, calculation_type (FLAT/PER_CONTAINER/PERCENT), value, currency, applies_to JSONB, valid_from, valid_to
cost_components	id, quote_version_id, component_code, description, amount, currency, source (RATE_CARD/PREDICTED/MANUAL), is_estimated
rate_history	id, lane_key, container_type, carrier_id, rate, currency, observed_at, source (partitioned monthly)
weather_risks	id, route_id, assessed_at, weather_condition, risk_score, delay_probability, storm_alerts JSONB, affected_legs JSONB, source, valid_until
shipment_risk_scores	id, shipment_id, quote_version_id, composite_score, factors JSONB (weather, congestion, customs, carrier, transit variance with weights), computed_at
alerts	id, shipment_id, quote_id, alert_type, severity (INFO/WARNING/CRITICAL), title, body, status (OPEN/ACK/RESOLVED), acknowledged_by, created_at
customs_requirements	id, shipment_id, country, direction (IMPORT/EXPORT), requirement_code, description, is_mandatory, compliance_status, regulation_citation_ids UUID[]
document_requirements	id, customs_requirement_id, document_type, is_provided, provided_file_key, notes
compliance_checks	id, shipment_id, agent_run_id, readiness_score, risk_flags JSONB, signed_off_by, signed_off_at, status
regulations	id, country, title, authority, regulation_ref, effective_from, source_url, last_verified_at
regulation_chunks	id, regulation_id, chunk_index, content, embedding VECTOR(1024), heading_path

hs_codes	code PK, description, chapter, restrictions JSONB, duty_notes
quote_approvals	id, quote_version_id, requested_by, approver_id, approver_role, threshold_breached, reason, decision (APPROVED/REJECTED), comment, decided_at
margin_policies	id, scope (GLOBAL/LANE/CUSTOMER_TIER/CARGO_TYPE), scope_key, floor_pct, target_pct, stretch_pct, is_active, effective_from
approval_rules	id, order_index, condition JSONB (margin below floor, discount above X, value above Y), approver_role, is_active
agent_runs	id, shipment_id, quote_id, workflow, status, started_at, finished_at, duration_ms, total_tokens, total_cost, error JSONB
agent_steps	id, run_id, agent_name, step_index, input JSONB, output JSONB, data_sources JSONB, model_version, prompt_version, tokens_in, tokens_out, latency_ms, status, error
integration_sync_logs	id, provider, operation, status, records_processed, error, started_at, finished_at
data_freshness	provider PK, last_success_at, last_attempt_at, staleness_minutes, threshold_minutes, is_stale
notifications	id, user_id, type, channel, title, body, link, is_read, sent_at
activity_logs	id, actor_id, action, entity_type, entity_id, before JSONB, after JSONB, ip, request_id, created_at (append-only, partitioned)
daily_quote_rollups	date, lane_key, carrier_id, customer_tier, quotes_generated, quotes_issued, quotes_accepted, avg_margin_pct, total_revenue
lane_performance_rollups	date, lane_key, volume, avg_transit_days, on_time_pct, avg_margin_pct, win_rate

6.3 Spatial & Vector Configuration

Parameter	Value / Policy
PostGIS usage	Ports as GEOGRAPHY (POINT,4326) with a GiST index; route geometry as LINESTRING. Enables great-circle distance, nearest-port lookup, and intersection tests against chokepoint and weather-hazard polygons.
Chokepoint layer	Static polygon table (Suez, Panama, Malacca, Hormuz, Bab-el-Mandeb, Gibraltar). Route intersection contributes directly to the risk score — this is cheap to compute and highly predictive.
pgvector	Regulation chunks, 1024 dimensions, HNSW cosine index, m=16, ef_construction=64.
Regulation chunking	Split by article/section heading, 400–700 tokens, 15% overlap, heading path prepended. Legal text degrades badly under naive fixed-size chunking — structure must be preserved.
Hybrid retrieval	Dense top-30 UNION full-text top-30, RRF fusion, cross-encoder rerank to top-5. Lexical matters because HS codes and regulation references are exact tokens.
Regulation freshness	Every regulation carries last_verified_at. Retrieval filters out records past the staleness threshold and the Customs Agent surfaces a warning rather than silently using outdated rules.

7. Authentication, Authorisation & Roles

7.1 Authentication Strategy

Aspect	Decision
Primary mechanism	JWT access (15 min) + rotating single-use refresh (7 days).
Token storage	Access token in memory; refresh in httpOnly, Secure, SameSite=Strict cookie. Never localStorage.
Two audiences	Internal staff and external customer-portal users authenticate through the same endpoint but receive different token audiences, land on different route trees, and are served by different serializers.
Organisation scoping	Every customer user carries a customer_id claim. All portal queries are filtered by it at the queryset level — the most likely serious bug in this system is one customer seeing another's quote.
Password policy	Argon2id, minimum 12 characters, breach-list check, lockout after 5 failures in 15 minutes.
MFA	TOTP, mandatory for PRICING_MANAGER, EXECUTIVE and ADMIN — the roles that can move money.
SSO	OIDC for internal staff; group claim mapped to role.
Service accounts	Scoped signed tokens for EDI ingestion and Core→Agent calls; quarterly rotation.

7.2 Role Definitions

Role	Description
CUSTOMER	External. Requests quotes, views own organisation's quotes at sell rate only, accepts or declines, uploads documents.
BROKER	Creates shipments, generates and builds quotes, adjusts margin within authority, issues approved quotes.
SENIOR_BROKER	Broker plus approval authority within a defined band and visibility across the team.
PRICING_MANAGER	Owns rate cards, margin policy, surcharge definitions; final approver on deep discounts.
OPERATIONS	Monitors routes, congestion, alerts and agent runs; manages master data.
COMPLIANCE_OFFICER	Owns customs validation, regulation corpus, and mandatory sign-off on compliance checks.
EXECUTIVE	Read-only across analytics, margins, forecast and lane performance.
ADMIN	Full configuration: users, roles, policies, approval rules, integrations, agent settings, audit.
SERVICE	Machine identity for EDI ingestion and inter-service calls. No UI access.

7.3 RBAC Permission Matrix

Capability	CUST	BRKR	SNR	PRC	OPS	CMPL	EXEC	ADM
Create shipment request	Yes	Yes	Yes	Yes	Yes	-	-	Yes
View own org quotes (sell only)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
View cost & margin	No	Yes	Yes	Yes	-	-	Yes	Yes
Generate quote (run agents)	Request	Yes	Yes	Yes	-	-	-	Yes
Adjust margin above floor	-	Yes	Yes	Yes	-	-	-	Yes
Adjust margin below floor	-	Request	Band 1	Any	-	-	-	Any
Approve / reject quote	-	-	Band 1	Any	-	-	-	Any
Issue quote to customer	-	Yes	Yes	Yes	-	-	-	Yes
Accept / decline quote	Yes	On behalf	On behalf	-	-	-	-	Yes
Manage rate cards / surcharges	-	-	-	Yes	-	-	-	Yes
Run customs check	-	Yes	Yes	-	Yes	Yes	-	Yes
Sign off compliance	-	-	-	-	-	Yes	-	Yes
Manage regulation corpus	-	-	-	-	-	Yes	-	Yes
Manage master data	-	-	-	Rates	Yes	-	-	Yes
View agent run traces	-	Own	Team	Team	All	Customs	Summary	All
Retry agent run	-	Own	Team	-	Yes	-	-	Yes
Executive analytics	-	Own	Team	Yes	Ops	Cmpl	Yes	Yes
Configure margin / approval policy	-	-	-	Yes	-	-	-	Yes
Configure integrations / agents	-	-	-	-	View	-	-	Yes

Manage users & roles	-	-	-	-	-	-	-	Yes
View audit log	-	-	-	Own	-	Own	-	Yes

The cost-visibility rule

Customer-role responses must have cost, margin, carrier buy-rate and win-probability fields removed

at the serializer level in the backend

, not hidden in the frontend. A dedicated `CustomerQuoteSerializer` is the only serializer a portal endpoint may use, and an automated test asserts that no forbidden key ever appears in a portal response payload. Hiding a field in React while the API still returns it is a data breach waiting for someone to open the network tab.

8. AI / ML & Agent Design

8.1 Division of Labour: Models vs LLMs vs Rules

This is the design decision that determines whether the system is trustworthy. Three distinct mechanisms, each used only where it is appropriate:

Mechanism	Used for	Never used for
Deterministic rules & arithmetic	Cost summation, surcharge application, margin floors, approval thresholds, FX conversion, validity expiry	Anything requiring judgement or interpretation of unstructured text
Classical ML models	Transit time, congestion forecast, market rate prediction, delay probability, win probability	Explaining themselves in prose; producing customer-facing text
LLM	Interpreting regulation text, reasoning over trade-offs, summarising risk, drafting rationale and narrative	Any arithmetic, any final price, any binding compliance determination

8.2 Agent Roster

Agent	Responsibility	Method & Data Sources
Route Agent	Identify viable routings, estimate transit time, compare carriers, assess port congestion, rank alternatives.	Graph search over the port/service network; transit-time model (LightGBM on lane, carrier, season, congestion, historical variance); congestion forecast from AIS and port data. LLM used only to explain the trade-off between the ranked options.
Pricing Agent	Build the buy-side cost: base rate, surcharges, origin/destination charges by incoterm; compare carrier pricing; flag anomalies.	Rate card lookup (deterministic) with a market-rate prediction model as a fallback when no contract rate covers the lane. Every component labelled RATE_CARD or PREDICTED so the broker knows what is firm.
Weather Agent	Assess route weather, identify storm and disruption risk, estimate delay probability, recommend alternatives.	Weather provider forecasts sampled along the route geometry; seasonal climatology; delay-probability classifier trained on historical weather-vs-actual-transit data.
Customs Agent	Validate import/export requirements, produce the documentation checklist, flag compliance risks, generate the readiness report.	RAG over the regulation corpus plus a deterministic rule table for common commodity/country pairs. Every claim must cite a regulation chunk or it is dropped. Output is advisory pending human sign-off.
Margin Agent	Recommend margin, analyse profitability, estimate win probability, produce a sensitivity view.	Constrained optimisation against the margin policy; win-probability model (features: lane, customer tier, margin, transit competitiveness, historical win rate, quote turnaround). Recommends only — the Core API enforces the floor.
Freight Intelligence Engine	Fuse all agent outputs, select the recommended routing, compute composite shipment risk, write the itemised rationale, flag contradictions.	Weighted scoring plus LLM narrative generation constrained to the structured facts produced upstream. Detects conflicts — e.g. the cheapest route is also the highest weather risk — and surfaces them rather than silently picking one.

8.3 Orchestration Graph

Node 1 — Route Agent (sequential). Must complete first; everything downstream is defined relative to a candidate routing. Produces the top 3 routings.
Nodes 2, 3, 4 — Pricing, Weather, Customs (parallel fan-out). Independent of one another and collectively the latency bottleneck, so they run concurrently. Each has its own timeout; a timeout degrades that dimension rather than failing the run.
Node 5 — Margin Agent (join). Requires the full cost build-up and the risk picture. Runs after the fan-out completes or times out.
Node 6 — Freight Intelligence Engine. Fuses, scores, writes rationale, flags contradictions.
Conditional edge. If the Route Agent finds no serviced port pair, the graph short-circuits to a "no routing available" result with an explanation — no point running pricing on a route that does not exist.
Checkpointing. State persisted after each node. Retry resumes from the last good checkpoint; external API cost is not re-incurred for completed nodes.

Degraded-mode matrix

Failure	System behaviour	User-visible effect
Weather provider unavailable	Fall back to seasonal climatology; mark the weather factor as low-confidence.	Quote proceeds; risk panel shows a degraded-data badge.
Commodity API returns null	Mark the commodity as unknown; flag the cost as low-confidence.	Quote proceeds; risk panel shows a degraded-data badge.

Carrier rate API unavailable	Use the most recent cached rate card; if stale beyond threshold, use the market-rate model and label every component PREDICTED.	Quote proceeds; broker sees an explicit "indicative pricing" banner and approval is forced.
Congestion data stale	Route Agent uses historical averages for that port.	Congestion gauge shows the data age.
Regulation corpus incomplete for the country	Customs Agent returns "insufficient coverage" rather than guessing.	Compliance check flagged for manual officer review. This is the correct behaviour, not a defect.
LLM provider down	Structured outputs still produced by models and rules; narrative rationale omitted.	Quote generates without the prose explanation.

8.4 ML Model Specifications

Model	Type	Key features	Target metric
Transit time estimator	LightGBM regression	Lane, carrier, service, container type, month, transshipment count, origin/destination congestion index, distance, historical variance for the lane	MAE $\leq$ 2.0 days; report a prediction interval, not a point estimate
Port congestion forecaster	Time series (Prophet / SARIMA per port)	Historical waiting hours, vessel arrival counts, seasonality, holidays	MAPE $\leq$ 20% at 7-day horizon
Market rate predictor	LightGBM regression	Lane, container type, week-of-year, fuel index, historical rate lags, capacity proxy, carrier	MAPE $\leq$ 12%; used only where no contract rate exists
Weather delay probability	Gradient-boosted classifier	Route weather severity samples, season, region, chokepoint crossings, historical delay outcomes	AUC $\geq$ 0.75; calibrated probabilities
Win probability	Logistic regression / LightGBM	Margin %, relative price vs lane median, transit competitiveness, customer tier, quote turnaround time, historical win rate for the customer	AUC $\geq$ 0.70; used for guidance, never to auto-adjust price
Composite risk score	Weighted rule model	Weather (0.25), congestion (0.20), customs (0.25), carrier reliability (0.15), transit variance (0.15)	Explainable by construction; every factor contribution shown in the UI

Cold-start reality

Several of these models need historical data the business may not have in usable form. Plan for it: transit time and market rate start as rule-based estimators (published schedules, rate card medians) with the ML model behind a feature flag, promoted only when a hold-out evaluation beats the rule baseline. Shipping a worse model than a simple average, and calling it AI, is the failure mode to avoid.

8.5 Guardrails

Control	Behaviour
Numeric guard	LLM output is parsed into a schema where all monetary and count fields are rejected. Numbers reaching a quote come only from rate cards, models or deterministic calculation. Enforced in code, tested explicitly.
Citation enforcement	Any customs requirement without a resolvable regulation citation is dropped from the output and logged as a coverage gap.
Staleness guard	Each data source has a maximum age. Exceeding it downgrades confidence, labels the output, and — for rates beyond the hard threshold — forces approval before issuance.
Margin floor enforcement	Applied in the Core API after the agent run, not inside the agent. An agent recommendation below floor is recorded and suppressed, never silently applied.
Contradiction detection	The Intelligence Engine flags internally inconsistent outputs (e.g. lowest-cost route carries the highest risk) for broker attention instead of resolving them silently.
Prompt-injection defence	Customer free-text (special requirements, cargo description) is delimited and labelled untrusted. It can inform, never instruct.
Cost control	Per-run token ceiling, daily budget, cached regulation retrieval, cheaper model tier for routine lanes.

8.6 Evaluation Harness

Golden set	Size	Metrics
Route set	200 lanes	Viable-routing rate, transit MAE, ranking agreement with SME judgement
Pricing set	150 historical quotes	Total cost deviation, component-level error, rate-card resolution correctness
Customs set	50 SME-reviewed	Checklist precision/recall, citation validity, false-confidence rate

Customs set	20000 reference scenarios	Outcomes: precision, recall, precision-recall, false confidence rate
Risk set	100 shipments with known outcomes	Risk score vs actual delay correlation, calibration
Safety set	40 adversarial inputs	Injection resistance, numeric-guard bypass attempts, stale-data handling

CI gate: merges touching agents/, ml/, prompts/ or optimization/ run the harness and are blocked on any headline metric regressing by more than 3%.

9. Third-Party Integrations

Integration	Purpose	Refresh	Implementation Notes
Carrier rate APIs / portals	Contract and spot rates, schedules	Daily + on-demand	One adapter per carrier behind a common interface. Where no API exists, support rate-card file import with a validation report before commit. Cache with a short TTL; every quote records which rate snapshot it used.
AIS / vessel tracking	Vessel positions, port calls, ETA signals	Hourly	Used to derive congestion and actual transit times for model training. Licensing terms constrain redistribution — never expose raw AIS to the customer portal.
Port data provider	Waiting hours, berth availability, terminal status	Every 6 hours	Feeds the congestion index and forecaster. Falls back to historical averages when unavailable.
Weather provider	Marine forecasts, storm warnings, wave height	Every 6 hours + on quote	Sampled at points along the route geometry rather than at endpoints only — endpoint-only sampling misses mid-ocean systems entirely.
Customs / HS-code reference	Tariff classifications, restrictions, documentation rules	Weekly + manual	Ingested into the regulation corpus with source URL and verification date. Compliance officer reviews and approves each ingestion batch.
FX rates	Multi-currency quotation	Daily	Rate frozen onto each quote version. Historical rates retained for reproducibility.
LLM provider	Reasoning, interpretation, narrative	On demand	Behind the LLMProvider interface with a configured failover provider.
Email (SMTP / transactional)	Quote delivery, approvals, alerts	Event-driven	Branded templates, PDF attachment or secure link, DKIM/SPF, bounce handling.
Teams / Slack	Approval requests, risk alerts	Event-driven	Incoming webhooks with actionable links back into the workbench.
Object storage	Quote PDFs, customer documents	—	Pre-signed URLs; server-side encryption; lifecycle archival.
Sentry / Langfuse	Errors and agent tracing	Continuous	Correlated by request ID across all three services.

Data-freshness monitoring is a first-class feature here

Unlike most integrations, these providers can fail *silently* — returning stale but well-formed data. A `data_freshness` table tracks last-success time per provider against a threshold, the Integrations page surfaces it, alerts fire on breach, and every quote records the age of the data behind it. A broker quoting Trans-Pacific rates from last week's snapshot without knowing it is the most expensive kind of bug this system can have.

10. End-to-End Project Workflow

10.1 Operational Flow

Enquiry — Broker or customer submits shipment details. Validated against port, commodity and container master data. Persisted as <code>ENQUIRY</code> .
Agent run — Route Agent produces candidate routings; Pricing, Weather and Customs run in parallel; Margin Agent joins; Intelligence Engine fuses. Progress streams live to the quote builder.
Business rules — Core API resolves the applicable margin policy, applies the floor, freezes FX and rate-card references, and determines whether approval is required.
Quote assembly — Version 1 created with cost breakdown, sell rate, transit, risk score, customs readiness, assumptions and itemised rationale. Status <code>DRAFT</code> or <code>PENDING_APPROVAL</code> .
Broker review — Broker inspects the agent rationale, may select an alternative routing, adjust surcharges or move the margin within authority. Any change creates a new version.
Compliance — If the customs readiness score is below threshold or the commodity is restricted, a compliance officer sign-off is mandatory before issuance.
Approval — If a threshold is breached, the quote routes to the approver resolved by the rule set. Approve, reject with reason, or request revision.
Issuance — PDF generated, emailed to the customer, published on the portal. Validity timer starts. Status <code>ISSUED</code> .
Outcome — Customer accepts, declines or lets it expire. All three outcomes are recorded and feed the win-probability model and lane analytics.
Monitoring — For accepted shipments, weather and congestion alerts continue until the shipment date, notifying the broker of material changes.

10.2 State Machines

Shipment

State	Allowed transitions	Trigger
ENQUIRY	→ QUOTING, CANCELLED	Quote generation starts
QUOTING	→ QUOTED, QUOTE_FAILED	Agent run completes or fails
QUOTED	→ WON, LOST, EXPIRED	Customer decision or validity lapse
QUOTE_FAILED	→ QUOTING	Retry after remediation
WON	→ BOOKED (Phase 2), CLOSED	Acceptance
LOST / EXPIRED / CANCELLED	Terminal	—

Quote

State	Allowed transitions	Trigger
DRAFT	→ PENDING_APPROVAL, APPROVED, SUPERSEDED	Submit, or auto-approve when within authority
PENDING_APPROVAL	→ APPROVED, REJECTED, SUPERSEDED	Approver decision
REJECTED	→ DRAFT (new version)	Broker revises
APPROVED	→ ISSUED, SUPERSEDED	Broker issues to customer
ISSUED	→ ACCEPTED, DECLINED, EXPIRED, SUPERSEDED	Customer action or expiry sweep
ACCEPTED / DECLINED / EXPIRED / SUPERSEDED	Terminal	—

Any edit to an `ISSUED` quote creates a new version and marks the old one `SUPERSEDED` . An issued price is never mutated — that record may be needed in a dispute.

10.3 Approval Rule Example

Order	Condition	Approver

1	Margin below the lane floor by more than 5 percentage points	PRICING_MANAGER
2	Margin below the lane floor by up to 5 percentage points	SENIOR_BROKER
3	Quote value above the configured high-value threshold	PRICING_MANAGER
4	Any component priced as PREDICTED because no contract rate exists	SENIOR_BROKER
5	Customs readiness score below threshold, or restricted commodity	COMPLIANCE_OFFICER (blocking)
6	New customer with no credit profile	SENIOR_BROKER

Rules evaluate in order; all matching rules apply, and the quote requires every resulting approval. Rules are data, editable by Admin, versioned, and every evaluation is written to the audit log.

11. Development Phases — Step by Step

Week 0 — Mobilisation & Foundations (pre-milestone)

Goal: the skeleton runs end to end and the domain is agreed before anyone writes a feature.

Track	Tasks
Repository & process	Repos created, branch protection, PR template, CODEOWNERS, conventional commits, ESLint import-boundary rule for pricing/internal.
Environment	Docker Compose: Postgres with PostGIS and pgvector, Redis, Core API, Agent Service, web, MinIO, Mailhog. One command to a running system.
Domain alignment	Half-day session with a freight SME. Agree: incoterm cost-responsibility matrix, surcharge codes in scope, container types, lane naming convention, margin policy structure. Write it down. Most defects in logistics software trace back to a domain assumption nobody wrote down.
Contracts	Freeze enums (statuses, cargo types, surcharge codes, roles). Publish the OpenAPI stub for frontend mocking.
Data	Acquire: port master (UN/LOCODE with coordinates), carrier list, at least one real rate card, 12–24 months of historical quotes with outcomes, historical transit actuals. Critical path.
Quality gates	Pre-commit hooks, CI running lint and tests, coverage floor 60%.
Deliverable	Running skeleton, seeded master data, agreed domain glossary, published API contract, CI green.

Milestone 1 — Weeks 1-2 — Route Intelligence & Quotation Foundation

Target: route recommendations for ≥ 98% of test lanes; transit MAE ≤ 2 days; quotation dashboard functional.

Track	Tasks
Backend	Accounts with JWT, roles and the two token audiences. Master data: ports with PostGIS geometry, carriers, services, commodities, container types, FX. Customers and tiers. Shipments app: model, conditional validation, state machine, CRUD, filters. Routing app: route and leg models, persistence of agent options. Skeleton quotes app so a routing can be attached to a draft. Audit middleware. Celery with the quotes and market-data queues.
Agent Service	FastAPI skeleton, service auth, structured logging, typed state. Port/service network graph and route search. Transit-time baseline estimator (rule-based) plus the LightGBM model trained on historical actuals; evaluate both, ship the better one. Congestion index from port data. Route Agent with ranking and an LLM-generated trade-off explanation. /v1/route/options. Route golden set and evaluation script.
Frontend	App shell, dual route trees, layouts, role guards, login. Design tokens and ui/ primitives. <Money /> and unit helpers. Shipment form with conditional cargo/hazmat sections and Zod validation. Shipment list and detail. Route Intelligence page with MapLibre route rendering, port markers, transit bars, carrier comparison table. Port congestion page. Quotation dashboard shell.
Data / ML	Clean and load port master. Build the lane key convention. Assemble the transit-time training set and hold-out split. Baseline vs model evaluation report.
QA	Unit tests on validation and the state machine. Contract tests. Playwright: login → create shipment → see route options on the map.
Definition of Done	A shipment entered through the UI returns ranked routings with transit estimates rendered on a map, persisted and filterable. Transit MAE evidenced on the hold-out set. Route coverage measured across 200 test lanes.

Milestone 2 — Weeks 3-4 — Pricing Intelligence & Margin Optimisation

Target: cost deviation ≤ 8% vs historical quotes; zero floor violations; quote in < 60 s p95.

Track	Tasks
Backend	Pricing app: rate cards, lines, surcharges, cost components, rate history. Rate card import with a pre-commit validation report. Deterministic cost build-up honouring incoterm responsibility. core/money.py with Decimal arithmetic and the rounding policy. Quotes app in full: versioning with immutability, line items, margin policy resolution, floor enforcement, approval rule engine, state machine, PDF generation, expiry sweep. Approval queue endpoints.
Agent Service	Pricing Agent: rate-card resolution, surcharge application, anomaly flagging, component source labelling. Market rate prediction model (baseline median vs LightGBM, ship the winner). Margin Agent: constrained optimisation against policy, win-probability model, sensitivity curve. Numeric guard implemented and unit-tested. /v1/pricing/estimate, /v1/margin/optimize. Pricing golden set of 150 historical quotes wired into CI.
Frontend	Quote Builder: route selection, cost breakdown table, surcharge editor, margin slider with live sell rate and floor warning, win-probability meter, agent rationale rail with SSE progress. Quote detail with version timeline, assumptions panel and PDF preview. Quote list with margin columns. Approval queue with breach reasons. Rate card management and import UI. Margin analysis dashboard. Customer portal quote view built and verified to contain no cost fields.
QA	Reconciliation harness against 150 historical quotes. Floor-violation test across every lane and customer tier. Decimal precision tests. Automated assertion that portal responses never contain cost, margin or buy-rate keys. Timing test for the p95 target.
Definition of Done	A shipment produces a complete priced quote with itemised cost, enforced margin, approval routing where required, and a branded PDF. Cost deviation and floor-violation targets evidenced.

Milestone 3 — Weeks 5-6 — Weather Risk & Customs Intelligence

Target: 100% route weather coverage; $\geq 90\%$ customs checklist accuracy; every quote risk-scored and explainable.

Track	Tasks
Backend	Risk app: weather risk records, composite shipment risk scoring with factor contributions, alerts with acknowledge/resolve, alert subscriptions. Customs app: requirements, document requirements, compliance checks, sign-off, regulation and chunk models, HS code reference. Blocking sign-off rule wired into the quote state machine. Weather, port and customs-reference integration adapters with sync logs and the data_freshness table. Notification dispatcher with email and Teams/Slack channels.
Agent Service	Weather Agent: route geometry sampling, storm risk assessment, delay-probability classifier, alternative routing advice. Customs Agent: regulation corpus ingestion, chunking, embedding, hybrid retrieval with rerank, requirement extraction with mandatory citations, readiness scoring, insufficient-coverage behaviour. Citation enforcer and staleness guard. Chokepoint intersection scoring using PostGIS. Parallel fan-out orchestration for nodes 2-4. /v1/weather/assess, /v1/customs/validate, /v1/regulations/search. Customs and risk golden sets.
Frontend	Risk dashboard with regional distribution and factor breakdown. Weather risk map overlay, storm alerts, delay probability chart. Alerts inbox. Customs dashboard, compliance check page with citation-linked checklist and officer sign-off action, document checklist with upload, regulation library with semantic search and freshness indicators. Risk and customs panels integrated into the quote builder and detail pages.
QA	SME review of 50 customs scenarios with a scoring rubric. Weather coverage test across all active lanes. Degraded-mode tests: kill each provider in turn and assert the documented fallback behaviour. Staleness-guard tests.
Definition of Done	Every generated quote carries a weather assessment, a customs readiness score with cited requirements, and a composite risk score with visible factor contributions. Compliance sign-off blocks issuance where required. SME accuracy target evidenced.

Milestone 4 — Weeks 7-8 — Freight Intelligence Engine & Enterprise Deployment

Target: $\geq 95\%$ agent-run completion; all five dashboards delivered; production live.

Track	Tasks
Backend	Analytics app: rollup models, nightly aggregation, KPI / lane / forecast / agent endpoints, CSV and PDF export as async jobs. Revenue forecast from weighted pipeline. Query optimisation via EXPLAIN ANALYZE on the slowest endpoints; N+1 elimination; Redis caching on dashboard reads. Audit log partitioning. Retention jobs.
Agent Service	Freight Intelligence Engine: weighted fusion, recommended-routing selection, contradiction detection, itemised rationale generation, composite confidence. Full orchestration with checkpointing, per-node timeouts and retry-from-checkpoint. Latency optimisation: provider response caching, parallel tuning, prompt compression. Final evaluation across all five golden sets.
Frontend	Executive dashboard, revenue forecast, lane performance. Agent monitor and run detail with full trace. Admin console: users, roles, margin policy, approval rules, agent settings, integrations health with freshness indicators, audit log. Accessibility pass. Bundle optimisation with route-level splitting and lazy map/chart loading. Empty, loading, error and degraded-data states audited on every page.
DevOps	Production Kubernetes manifests / Helm chart. Rolling deploy with health gates. Prometheus, Grafana dashboards including per-provider freshness panels, alert rules. Backups with a tested restore. Secrets to vault. Load test at 3x expected peak.
QA & UAT	Full regression. Security review: OWASP Top 10, dependency and container scanning, penetration test focused on cross-customer data access. Five-day UAT with real brokers and a compliance officer. Bug triage and fix cycle.
Documentation	Broker user guide, compliance officer guide, admin guide, runbook (provider outages, stale data, failed runs, restore procedure), API reference, architecture decision records, recorded handover.
Definition of Done	Platform live, all five dashboards delivered, orchestration completion rate evidenced over 200 runs, monitoring and alerting active, restore verified, documentation handed over, no open critical or high defects.

12. Deployment Architecture

12.1 Environments

Environment	Purpose	Data	Deploy trigger
Local	Development	Seeded master data, synthetic quotes, mocked providers	Docker Compose, manual
Dev	Shared integration	Synthetic + sandbox provider keys	Auto on merge to develop
Staging	UAT, load & security testing	Anonymised production-like, real read-only provider keys	Auto on merge to release/*
Production	Live	Real	Manual approval on tagged release

12.2 Runtime Topology (production)

Component	Deployment	Scaling & notes
Web (static SPA)	CDN + object storage	Immutable hashed assets; map tiles served from the provider CDN.
Ingress	Nginx / cloud LB	TLS 1.3, HTTP/2, rate limiting, WAF, security headers. Separate host or path prefix for the customer portal.
Core API	3+ pods (HPA)	Gunicorn + Uvicorn workers, readiness/liveness probes, graceful shutdown.
Agent Service	2+ pods (HPA on queue depth & latency)	Higher memory and CPU limits; separate node pool. Model artifacts pulled at startup and cached on a volume.
Celery workers	Per-queue deployments	quotes scales on backlog; market-data runs at fixed low concurrency to respect provider rate limits and licensing quotas.
Celery Beat	Single replica, leader-elected	Duplicate schedules would double-charge metered provider APIs.
PostgreSQL	Managed + read replica	PostGIS and pgvector on primary; analytics and dashboard reads served from the replica. PITR enabled.
Redis	Managed, replicated	Separate logical DBs for broker, cache and rate limiting.
Object storage	S3 bucket	Private, encrypted, lifecycle rules. Quote PDFs retained per the contractual retention period.
Observability	Prometheus, Grafana, Loki, Sentry, Langfuse	Includes a dedicated data-freshness dashboard per provider.

12.3 CI/CD Pipeline

1. On pull request — lint → type check → unit tests → coverage gate → dependency and secret scan → build images → container vulnerability scan.
2. Domain safety gate — run the money-precision suite and the portal field-leakage assertion on every build. These never get skipped.
3. AI gate — if agents/ , ml/ , prompts/ or optimization/ changed, run the evaluation harness; block on > 3% regression in any headline metric.
4. On merge to develop → push images → deploy Dev → smoke tests → notify.
5. On release branch → deploy Staging → full E2E and load test → UAT window opens.
6. On tagged release → manual approval → backwards-compatible migration job → rolling deploy with health gates → smoke tests → auto-rollback on failure.

Migration rule

Expand first, contract later. Migrations must be compatible with the previous application version, since old and new pods run concurrently during a rolling deploy. Never drop or rename a column in the same release as the code change that stops using it — and never backfill a large historical table inside a migration; use a separate reversible data job.

12.4 Reliability & Recovery

Concern	Policy
Backups	Daily full plus continuous WAL archiving. Restore drill executed and documented before go-live. Quote data is contractually significant — loss is not recoverable by regeneration, since market data has moved on.

RPO / RTO	RPO 15 minutes, RTO 2 hours, with the recovery sequence written in the runbook.
Degraded modes	As specified in section 8.3. Every degradation is visible in the UI, never silent.
Alerting	Page on: API 5xx > 2%, agent-run failure > 10%, any provider stale beyond threshold, quote-generation p95 > 90 s, queue depth sustained > 300, LLM cost above daily budget, database connections > 80%.
Feature flags	Each agent, each ML model (vs its rule baseline), and each integration individually toggleable at runtime. A failing model must be switchable off in seconds, not in a deploy cycle.

13. Non-Functional Requirements, Testing & Security

13.1 Non-Functional Targets

Attribute	Target	Verification
API latency (CRUD)	p95 < 300 ms	Load test, Prometheus histograms
Quote generation (full run)	p95 < 60 s, p50 < 25 s	End-to-end timing under 10 concurrent runs
Route options only	p95 < 8 s	Used for the live estimate on the enquiry form
Dashboard load	p95 < 1.5 s	Served from rollups and the read replica, never live aggregation
Throughput	200 quotes/day sustained, 50/hour peak	Load test at 3× peak
Availability	99.5% monthly	Uptime monitoring; provider degradation does not count as downtime
Monetary accuracy	Exact to 4 decimal places, no float anywhere	Dedicated precision test suite in CI
Quote reproducibility	Any historical quote regenerates identically from frozen inputs	Automated reproduction test over a sample of past quotes
Frontend load	LCP < 2.5 s; map bundle lazy-loaded	Lighthouse in CI
Accessibility	WCAG 2.1 AA	axe scan + manual keyboard pass
Data retention	Quotes 7 years, agent traces 1 year, market data 3 years	Retention jobs with documented policy

13.2 Testing Strategy

Level	Tooling	Coverage expectation
Unit	pytest, Vitest	≥ 80% on services, margin policy, approval rules, cost build-up, state machines, guardrails.
Money precision	pytest with Decimal fixtures	Every calculation path, every rounding boundary, multi-currency conversion.
Authorisation	pytest parametrised by role	Every endpoint × every role. Plus the portal field-leakage assertion.
Component	React Testing Library	All shared components and forms; <Money /> formatting cases.
API contract	pytest + schemathesis	Every endpoint against its OpenAPI schema, including error shapes.
Integration	pytest + testcontainers	Real Postgres with PostGIS and pgvector. Providers mocked at the adapter boundary with recorded fixtures.
AI evaluation	Custom harness	Five golden sets in CI on every AI-layer change.
Degraded-mode	Integration suite	Each provider failure scenario asserts the documented fallback.
E2E	Playwright	Broker quote journey, approval journey, compliance sign-off, customer portal journey.
Load	k6 / Locust	Sustained and spike profiles before each release.
Security	OWASP ZAP, bandit, npm audit, Trivy	Automated in CI; manual penetration test before go-live.

13.3 Security Controls

- **Tenant isolation:** every customer-scoped queryset filtered by `customer_id` from the token, never from a request parameter. IDOR tests for every resource-by-id endpoint are mandatory, not optional.
- **Commercial confidentiality:** cost, margin and carrier buy-rates stripped at the serializer for customer-facing responses and excluded from any customer-facing PDF, email or export.
- **Transport:** TLS everywhere including service-to-service. HSTS, CSP, X-Frame-Options set at the edge.
- **Input:** every payload validated by serializer or Pydantic model; parameterised queries only; uploaded documents type-checked, size-capped and virus-scanned.
- **Secrets:** vault-managed, rotated quarterly, CI fails on any secret detected in a diff. Carrier and data-provider credentials are commercially sensitive and access-logged.
- **Provider licensing:** AIS and rate data carry redistribution restrictions. Export and API responses respect them; this is a contractual obligation, not a preference.
- **Audit:** append-only, immutable log of every price change, margin override, approval decision and compliance sign-off, with actor, request ID, and before/after state.

14. Team Structure, Ways of Working & Risk Register

14.1 Suggested Squad Split

Squad	Size	Ownership
Platform & Core API	3-4	Auth, shipments, quotes, versioning, margin policy, approval engine, audit, migrations.
Agents & ML	3-4	All five agents, Intelligence Engine, ML models, optimisation, guardrails, evaluation harness.
Frontend	3-4	Workbench, portal, five dashboards, maps, design system, accessibility.
Integrations & Data	2-3	Carrier, AIS, weather, port, customs and FX adapters; freshness monitoring; master data; dataset curation.
QA & DevOps	2	CI/CD, environments, test automation, monitoring, security scanning, releases.
Domain SME (part-time)	1	Freight expertise: validates cost logic, customs outputs, route plausibility. Not optional — a team without domain review will ship confidently wrong logistics.

14.2 Ways of Working

Practice	Standard
Branching	main ← release/* ← develop ← feature/FQ-123-short-slug. Hotfixes from main, merged back to both.
Commits	Conventional commits driving automated changelog.
Pull requests	~400 lines maximum, two approvals, CI green. Any PR touching money, margin or approval logic requires a review from the Platform squad lead specifically.
Sprints	One-week sprints inside each two-week milestone; Friday demo against the milestone Definition of Done.
SME review	Every milestone includes a scheduled domain review session. Agent outputs are checked by a human who knows freight before the metric is accepted.
Documentation	ADR for every significant decision (why this risk weighting, why this chunking strategy). Domain glossary maintained as a living document.
Definition of Ready	Acceptance criteria, API contract if crossing a service boundary, test approach, and — for domain logic — a worked example agreed with the SME.

14.3 Risk Register

Risk	Impact	Likelihood	Mitigation
Carrier rate data unavailable or not machine-readable	High	High	Assume file-based rate cards as the primary path from Week 0; treat API access as an enhancement. Build the import-and-validate flow in Milestone 2 regardless of what integrations exist. This is the single most common blocker in freight-tech builds.
Insufficient historical data for ML models	High	High	Rule-based baselines behind feature flags; ML promoted only when it beats the baseline on hold-out. Ship a working product with simple estimators rather than a delayed one with sophisticated ones.
Customs regulation corpus incomplete or outdated	High	High	Scope Phase 1 to a defined set of trade lanes and countries agreed with the compliance officer. Mandatory human sign-off. "Insufficient coverage" is a valid, visible output. Never let the model fill a gap with plausible text.
External provider cost exceeding budget	Medium	Medium	Aggressive caching, scheduled batch refresh rather than per-quote calls where possible, quote-generation rate limits per user, cost dashboard from Milestone 2.
Floating-point or rounding errors in pricing	High	Medium	Decimal end to end, a dedicated precision test suite in CI from Milestone 2, single rounding policy in one module. A one-cent discrepancy destroys broker trust faster than a missing feature.
Cross-customer data exposure via the portal	Critical	Low	Token-derived scoping, dedicated customer serializers, automated field-leakage assertion in CI, IDOR tests, penetration test focused on this specific vector.
Agent latency making the tool feel slow	Medium	Medium	Parallel fan-out, per-node timeouts with degradation, live progress streaming so waiting feels productive, cached provider responses, fast route-only estimate on the enquiry form.
Broker distrust of AI-generated pricing	High	Medium	Every number traceable to its source; component-level labelling of firm vs predicted; full rationale visible; broker always has final control of margin. Involve two brokers in UAT from Milestone 2 onward. Transparency is the adoption strategy.
Scope creep — booking, tracking, invoicing	High	High	Out-of-scope list is contractual. New requests go to Phase 2 by default; only the sponsor can promote one, and only by removing something equivalent.

14.4 Post-Launch Roadmap (indicative)

- **Phase 2:** Carrier booking integration; shipment tracking against the quoted routing; automated rate-card refresh; multimodal pre- and on-carriage legs.
- **Phase 3:** Contract negotiation support with counter-offer modelling; customer-specific pricing agreements; automated tender responses.
- **Phase 4:** Predictive disruption alerts across the live shipment portfolio; carbon emissions estimation per routing; multi-tenant deployment for brokerage networks.